

Education

B.S. Stevens Institute of Technology, Hoboken, NJ *Quantitative Finance* | GPA: 3.7 / 4.0

Anticipated May 2028

- *Awards:* Edwin Augustus Stevens Scholarship (\$104k), Presidential Scholarship (\$20k)
- *Coursework:* Financial Time Series, Algorithms (C++), Advanced Probability & Stochastic Processes for QF, Financial Econometrics, Advanced Time Series & Machine Learning, Market Microstructure Trading Strategies, Linear Algebra

Projects

Polymarket US Quantitative Trading System [Paper](#) [🔗](#)

Jun 2026 – Jul 2026

- Built two strategies trading Polymarket US World Cup player-prop contracts, earning \$815 profit on \$1,200 (+67.9%) in 35 days
- Engineered an unattended, Ed25519-authenticated Python pipeline that logged 157,595 three-leg quotes to SQLite and recomputed each player's fair value up to every 60 seconds before kickoff; traded its signals by hand, bidding below fair value
- Validated the core mispricing with a Monte Carlo null-hypothesis test: 71.4% of positions reverted toward fair value vs. 18.8% under a volatility-matched random walk ($p = 0.00003$); the patience strategy earned \$274 over 40 positions
- Ran an event-driven speed strategy that bought contracts at \$0.92–\$0.99 and won all 136 held positions (+\$541)

Vanguard ETF Challenge [Article](#) [🔗](#)

Sep 2025 – Dec 2025

- Ranked #1 in total return (+2.58%) and 2nd overall by Sharpe ratio (1.49) among 50 teams in an 8-week Vanguard ETF competition, the only team to beat the S&P 500
- Analyzed win rates across the team's millions of simulated efficient-frontier portfolios to guard against early losses in the 8-week window
- Designed and pitched 'RHO,' an 11-asset, 7-sector ETF screened to under 0.574 correlation with the S&P 500; backtested a 0.84 Sharpe vs. the S&P 500's 0.52 at 12.5% volatility, placing 2nd of 6 teams in the NYSE pitch to Vanguard

10-Q Equity Research Agent [Repo](#) [🔗](#) | [Live App](#) [🔗](#)

Summer 2025 – Present

- Built a 6-step LLM pipeline (Python, Gemini API) that turns each company's 10-Q text and XBRL financials, pulled live from SEC EDGAR, into a Bullish/Neutral/Bearish call; deployed as a public Streamlit app
- Backtested 500+ S&P 500 filings (growing nightly) on point-in-time data only, benchmarking against Wall Street consensus, momentum, and always-bullish baselines with binomial tests and a long-short portfolio
- Found the agent matched Wall Street at picking S&P 500 outperformers (51% vs. 50%), with neither beating a coin flip, and its 54% raw hit rate trailed a 64% always-bullish baseline ($p = 0.07$)

Kaiser-Stats [Repo](#) [🔗](#) | [Live App](#) [🔗](#)

Jul 2026 – Present

- Built a stats engine unifying 5 seasons (2022–2026, ~280+ games) of a local soccer league's inconsistent spreadsheet history into one per-player dataset
- Automated weekly updates: a scheduled routine uses the Gemini API to parse the league's game-report emails and add the weekend's games, then a separate stress tester independently validates the results
- Shipped a public registration and stats website now used by a 110+ person league mailing list, backed by data-integrity safeguards (player identity resolution, plus-minus sanity checks) and a transparent power-ranking formula (Next.js, Supabase)

FIBA Annual Stock Pitch Competition [Presentation](#) [🔗](#)

Sep 2025 – Oct 2025

- Placed 3rd of 15+ teams pitching a DCF valuation model to evaluate Disney (NYSE: DIS), covering competitive advantage, headwinds, and tailwinds to judges

Skills

- *Programming:* Python (Pandas, NumPy, scikit-learn), SQL, C++, TypeScript/JavaScript, R, Git
- *Involvements:* Finance Investment Banking & Advisory Society (FIBA), Scholars of Finance (SoF), Poker Club
- *Data/ML:* statistical modeling, predictive analytics, reinforcement learning, natural language processing, time-series forecasting, optimization, stochastic processes, Monte Carlo methods, portfolio optimization, risk management